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11/02/26

Experiment-5: Disease Prediction using Naive Bayes and Neural Network with Comparison of Classifiers

ML LAB

PID - 246001

In [3]:
```
import time
import pandas as pd

from sklearn.datasets import fetch_california_housing
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
```

In [5]:
```
data = fetch_california_housing()
print(data.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
```

In [6]:
```
X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

print(X.head())
print("Smaples:", len(X))
print(y[:10])
```

```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude
0    -122.23
1    -122.22
2    -122.24
3    -122.25
4    -122.25
Smaples: 20640
[4.526 3.585 3.521 3.413 3.422 2.697 2.992 2.414 2.267 2.611]
```

In [7]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X,y,test_size=0.2,random_state=42
)
```

In [15]:
```
tree = DecisionTreeRegressor()

start = time.perf_counter()
tree.fit(X_train, y_train)
tree_time = time.perf_counter() - start

pred_tree_test=tree.predict(X_test)
err_tree_test = mean_absolute_error(y_test, pred_tree_test)
pred_tree_train = tree.predict(X_train)
err_tree_train = mean_absolute_error(y_train, pred_tree_train)

print("Decision Tree Train MAE:", err_tree_train)
print("Decision Tree Train MAE:", err_tree_test)
print("Decision Tree Train Time:", tree_time)
```

```
Decision Tree Train MAE: 4.206368242523607e-17
Decision Tree Train MAE: 0.45272931928294574
Decision Tree Train Time: 0.23090720000004694
```

In [18]:
```
rf = RandomForestRegressor(n_estimators=100)

start = time.perf_counter()
rf.fit(X_train, y_train)
rf_time = time.perf_counter() - start

pred_rf_test = rf.predict(X_test)
err_rf_test = mean_absolute_error(y_test, pred_rf_test)
pred_rf_train = rf.predict(X_train)
err_rf_train = mean_absolute_error(y_train, pred_rf_train)

print("Random Forest Regressor train MAE:", err_rf_train)
print("Random Forest Regressor Train MAE:", err_rf_test)
print("Random Forest Regressor Time:", rf_time)
```

```
Random Forest Regressor train MAE: 0.12137927641715157
Random Forest Regressor Train MAE: 0.32745610251938
Random Forest Regressor Time: 14.637242900000047
```

In [32]:
```
gb = GradientBoostingRegressor(n_estimators=300, learning_rate=0.1,max_depth=5)

start = time.perf_counter()
gb.fit(X_train, y_train)
gb_time = time.perf_counter() - start

pred_gb_test = gb.predict(X_test)
err_gb_test = mean_absolute_error(y_test, pred_gb_test)
pred_gb_train = gb.predict(X_train)
err_gb_train = mean_absolute_error(y_train, pred_gb_train)

print("Gradient Boosting Regressor train MAE:", err_gb_train)
print("Gradient Boosting Regressor Train MAE:", err_gb_test)
print("Gradient Boosting Regressor Time:", gb_time)
```

```
Gradient Boosting Regressor train MAE: 0.22968324793032216
Gradient Boosting Regressor Train MAE: 0.30703343451110754
Gradient Boosting Regressor Time: 19.71050830000013
```

In [33]:
```
print("\n ==== Comparison For Test Mae ==== \n")
print("Decision Tree Test:",err_tree_test)
print("Random Forest Test:",err_rf_test)
print("Gradient Boost Test:",err_gb_test)
```

```
 ==== Comparison For Test Mae ====

Decision Tree Test: 0.45272931928294574
Random Forest Test: 0.32745610251938
Gradient Boost Test: 0.30703343451110754
```

In [34]:
```
print("\n ==== Comparison For Train Mae ==== \n")
print("Decision Tree Train:",err_tree_train)
print("Random Forest Train:",err_rf_train)
print("Gradient Boost Train:",err_gb_train)
```

```
 ==== Comparison For Train Mae ====

Decision Tree Train: 4.206368242523607e-17
Random Forest Train: 0.12137927641715157
Gradient Boost Train: 0.22968324793032216
```

In [38]:
```
print("\n ==== Comparison For Time Mae ==== \n")
print("Decision Tree Time:", tree_time)
print("Random Forest Time:", rf_time)
print("Gradient Boost Time:", gb_time)
```

```
 ==== Comparison For Time Mae ====

Decision Tree Time: 0.23090720000004694
Random Forest Time: 14.637242900000047
Gradient Boost Time: 19.71050830000013
```